

The Recycling Feature

Complete reference: research findings, verified partner directory, the RVRT verification layer, the contamination & sorting guidance spec, and the full feature roadmap

This document consolidates everything gathered so far on KaPhor's recycling feature into one reference: the statistics and structural problems from the textile-recycling research briefing, the RVRT (Recycler Verification & Risk-Weighted Route-Tracking) patentable extension to GLIE, a directory of verified partner addresses, the Contamination & Sorting Guidance spec, and a prioritised roadmap of every feature idea discussed for this part of the app.

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PART 1

Research Briefing

Statistics and structural problems in India's textile recycling system

1.1 Headline Statistics

Metric	Figure	Source
India's annual textile waste	~7.7–7.8 million tonnes/year (~8.5% of world total)	Fashion for Good, CSTEP, Landbell
Pre-consumer waste recovery rate	~95–97% recovered	Fashion for Good / CSTEP
Post-consumer waste recycled	~1,155 KTPA out of ~4,100 KTPA generated	Fashion for Good / CSTEP composite
Reused or recycled (all channels)	59% of total waste generated	Fashion for Good
Re-enters global supply chains as new raw material	Only 10–40% of the reused/recycled share	Fashion for Good
Domestic post-consumer waste to landfill	43%, vs. ~1% of pre-consumer/imported waste	Fashion for Good
Global textile waste reaching recycling facilities	Only 12–15%; of that, ~1% becomes new clothing	Regulatory-intervention paper (global, not India-specific)
Livelihoods sustained by India's informal recovery sector	4–4.5 million, mostly women from marginalised communities	Ministry of Textiles report, Mar 2026
Recovery rate achieved by informal sector	~55%, despite no formal recognition	Ministry of Textiles report, Mar 2026

Panipat, Haryana is the only Indian textile-recycling cluster with concrete, repeatedly-cited operating statistics — the de facto case study for what a recycling-center drop-in looks like at scale.

1.2 Where the Waste Comes From

Fashion for Good's breakdown of the ~7.8 million tonnes is the most granular composition data available: 42% pre-consumer, 58% post-consumer. Over 90% of nationally processed waste originates domestically — imports are a real flow but a minority share nationally, even though they dominate specific hubs like Panipat.

1.3 Problem A — Fragmentation & Opacity in the Supply Chain

There is no unified, centrally maintained directory of textile recycling centers, drop-off points, or processing facilities in India. Capacity is split across five non-overlapping, non-communicating data sources:

- **Industrial recycling clusters** — Panipat being the dominant, almost only well-documented example
- **EPR-registered intermediaries** — plastics, e-waste and batteries are tracked; textile-specific EPR registration counts were not found in any source reviewed
- **NGO/social-enterprise donation networks** — Goonj, Share At Door Step, Bombay Recycling Concern — which route wearable clothing to resale/redistribution rather than material recycling
- **Named startups & Textile Recovery Facilities** — Saahas Zero Waste, ReCircle, Respun, EcoDhaga — concentrated in Bengaluru, Gurugram, Delhi-NCR and Mumbai

- **The informal sector** — kabadiwalas, scrap dealers, the Waghri/Devipujak barter network — undocumented but handling a very large share of actual collection nationally

Practical consequence: neither a consumer nor a platform like KaPhor can currently determine, for any given garment, which channel it will pass through, what happens there, or how much of its mass is genuinely recycled versus downcycled, exported, or landfilled.

1.4 Problem B — Worker Health Harm, Concentrated in Import-Heavy Hubs

Independently confirmed across five outlets (CNN, The Guardian, ThePrint, Atmos, The Diplomat) reporting from Panipat between October 2025 and May 2026:

- COPD is the most common condition among textile workers per Panipat Civil Hospital, attributed to dust and fibre inhalation
- 93% of families near recycling clusters reported serious health issues — chronic respiratory disease, TB, cancer — within the past five years
- Workers in dyeing/bleaching — disproportionately women in early-stage sorting/cutting — handle acid and dyes without gloves; no on-site medical facilities
- Untreated wastewater from dyeing/bleaching flows through open drains, reportedly reaching the Yamuna River
- Much of the highest-risk material handled in Panipat is imported fast-fashion waste from the US, UK, Japan and Europe

Framing that matters for KaPhor: this is not “textile recycling inherently causes respiratory illness” — it is a specific, addressable failure of PPE, ventilation and enforcement at high-volume, import-heavy, weakly regulated facilities. That distinction is what makes a “verified facility” feature meaningful rather than performative.

PART 2

RVRT

Recycler Verification & Risk-Weighted Route-Tracking Layer — a patentable extension to GLIE's RECYCLE pathway

2.1 The Gap This Closes

GLIE's RECYCLE branch (GLIE < 0.40) currently does one thing: geo-query the recycler database and match by fiber type to the nearest facility. It applies no verification, safety, or sourcing criteria. Given that India's recycling sector is largely undocumented and that worker respiratory illness is concentrated at high-volume, import-heavy hubs like Panipat, routing blindly by distance risks sending KaPhor's highest-risk garments to the least safe, least accountable facilities.

RVRT replaces that single step with a sub-router, the same way GLIE's UPCYCLE branch already has its own RAG-retrieval sub-process. The novel mechanism: the weight given to a facility's safety profile is not fixed — it scales with how risky the specific garment is to process.

2.2 Formulas

ProcessingIntensityScore (PIS) — how risky this garment is to process. Reuses GLIE's existing ConditionScore and T2 fiber data — no new input pipeline needed.

FacilityTrustScore (FTS) — the facility's report card, built from verification tier, PPE/ventilation documentation, and effluent handling.

RouteScore — the adaptive combination and core novel mechanism. Facility-selection weights are computed per garment from PIS, not fixed:

$$\text{RouteScore} = W_d \cdot \text{Distance} + W_f \cdot \text{FTS} + W_c \cdot \text{CapabilityMatch} + W_m \cdot \text{SourceMatch}$$

where $W_d + W_f + W_c + W_m = 1.00$ always — consistent with GLIE's own top-level weight constraint, and the weights themselves shift with PIS (higher-risk garments push weight toward W_f).

2.3 Routing Logic & Verification Decay

Every routing decision logs the exact weights and scores that produced it (RouteDecisionRecord) — auditable, mirroring how GLIEAssessment already logs all four sub-scores. Verification tiers auto-downgrade after 180 days without re-check, so trust signals can't go stale silently. The user-facing badge shows at the same disposal screen GLIE already uses for carbon/water savings — no new screen required.

2.4 Dataset Schema — Table T6: Recycler Verification

Format: PostgreSQL/Prisma, joined to the existing recycler reference via facility_id. Extends T1–T5 in the same format. Launch target: 10–15 named partners, populated via a self-onboarding form.

2.5 API Addition

Extends **POST /api/glie/assess** — RECYCLE branch only. RESELL and UPCYCLE responses are unchanged.

2.6 Patentability Summary

Suggested patent title: "A computer-implemented method for risk-adaptive facility selection in automated textile end-of-life routing, wherein selection-criteria weighting is dynamically computed from a garment-specific processing-intensity estimate, combined with a time-decaying, tiered facility verification record."

Position RVRT as a dependent claim attached to the existing GLIE application rather than a separate filing — it reuses GLIE's ConditionScore as a direct input and applies the same weighted multi-factor philosophy to a second problem. Confirm this structure with the patent attorney; combining vs. separate filing is a legal call, not a technical one.

2.7 Next Steps

- Sense-check the PIS/RouteScore default weights with the team before locking them
- Onboard 5–10 named partners (Goonj, Saahas, Respun, EcoDhaga) via a self-reported form — populates T6 at SELF_REPORTED tier fast
- Defer automated photo-based safety scoring — start with manual review; a CV classifier needs labelled data not yet available
- Build the 180-day decay job early — cheap, and keeps the trust layer honest as the partner list grows
- Bring the RouteScore mechanism and patentability section to the next patent attorney conversation — the adaptive-weighting mechanism is the strongest individual claim

PART 3

Verified Partner Directory

Only entries with a confirmed street-level or precise-locality address

3.1 Goonj

Location	Address
Delhi (HQ / Processing Centre)	Rajesh Pilot Marg, Madan Pur Khadar, Near Sunder Public School, New Delhi – 110076
Mumbai (registered address)	1st Floor, Highway Rd, Kashmirira, Post Mira, Near Laxmi Car Showroom, Mira Road, Maharashtra – 401107
Bangalore (Office #1)	Sy. No. 51/1, Chikka Begur Gate, Near CPG BPO, Hosur Main Road, Kudlu Gate, Bangalore – 560068
Bangalore (Office #2)	Survey No. 25&26, 37/1, 6th Main, AECS Layout Block A, Singasandra, Bengaluru – 560068
Chennai	Siva Parvathy Kalayna Mandapam, Plot No. 25/26, Door No. 4/558, 2nd Main Rd, S. Kolathur, Viduthalai Nagar, Kovilambakkam, Chennai – 600117
Hyderabad	H.No. 7-1/8, Plot No. 8, Suncity Phase-2, Radha Nagar, Hyderabad – 500086
Dehradun	C-63, Nehru Colony, Dharampur, Near Fountain Chowk, Dehradun – 248001
Rishikesh	Near Lal Bahadur Shastri School, Dhalwala, Rishikesh, Tehri Garhwal – 249201
Kochi (Tripunithura dropping centre)	One Little Earth, Old Bus Stand, Opp. Central Cinemas, Near Tripunithura Sub Post Office, Kochi – 682301

3.2 Saahas Zero Waste — Textile Recovery Facility

Sy No. 21/3A, 21/3B, 21/3C, 21/3D, Kalena Agrahara, Bannerghatta Rd, Bengaluru

3.3 Hasiru Dala / Saamuhika Shakti / CAIF — Textile Recovery Center

Cheemasandra, Hoskote, Bengaluru, Karnataka – 560049

Waste-picker-empowerment pilot — a strong RVRT fit given its labour-focus mission.

3.4 Swachagraha Kalika Kendra

14th A Cross Rd, Sector 4, HSR Layout, Bengaluru, Karnataka – 560102

3.5 EcoDhaga

16/9C, 2nd Floor, 3rd Cross Road, Opposite Koramangala Police Station, 6th Block, Koramangala, Bengaluru, Karnataka – 560047

Resale/upcycling thrift operation, not fiber recycling — confirm which routing branch (RESELL/UPCYCLE vs. RECYCLE) this belongs to before onboarding into T6.

3.6 Not Included — No Verified Address Found

Organisation	What was found instead
ReCircle	Material Recovery Facility located in Dahisar, Mumbai – 400068 (pincode/locality confirmed across multiple press sources; exact street address not published)
Respun	Located in Panipat, Haryana — city confirmed, no street address found. Flagged in the research as sitting in the same import-heavy hub identified as the highest-risk zone for worker health
Bombay Recycling Concern	Domain (gobrc.in) and general Mumbai association found; no verifiable physical address located
Share At Door Step (SADS)	Bengaluru-headquartered, pickup-only model — no fixed public drop-off address by design

These four need direct outreach to confirm GPS-precise addresses, consistent with RVRT's own "self-onboarding form" next step rather than further web research.

PART 4

Contamination & Sorting Guidance

A point-of-disposal prep layer for the GLIE RECYCLE branch

4.1 Why This Matters

The Ministry of Textiles report attributes India's low post-consumer recycling rate directly to segregation being “economically unviable due to challenges in segregation, contamination, and lack of traceability” — not a shortage of recycling facilities. Today, KaPhor's RECYCLE branch hands a garment off to a facility and stops, doing nothing to reduce the chance it arrives wet, mixed-fiber, or carrying non-textile attachments. This feature intervenes at the one moment KaPhor actually controls: between the RECYCLE verdict and the user physically dropping the item off.

Root cause, not symptom: facility count is not the binding constraint on India's textile recycling rate — contamination and sorting failure is. This feature is prioritised over onboarding more facility partners for exactly that reason.

4.2 Where It Sits in the Flow

Step	Owner	Status
GLIE Routing Decision (score < 0.565 → RECYCLE)	GLIE Engine	Existing
Contamination & Prep Checklist	This feature	NEW
Facility Match (risk-adaptive selection)	RVRT Layer	Existing
Disposal screen — “This item will go to [Verified Partner]”	RVRT Layer	Existing

4.3 What the Checklist Contains

Generated dynamically per garment from data GLIE has already produced (T2 fiber data, ConditionScore damage tags) — not a static list.

Universal steps — every RECYCLE item

- Wash and fully dry the garment — damp items are rejected at intake or degrade during storage
- Remove non-textile attachments: buttons, zippers, metal hooks, sequins, rhinestones
- Cut off and discard care labels if they're a different fiber blend than the main garment

Fiber-specific steps — pulled from T2 material data

- **Blended fabrics** (cotton-lycra, poly-cotton): flagged as “mixed fiber” — routes to a different facility tier than pure cotton
- **Denim:** remove leather patches/rivets if detachable
- **Silk/wool:** keep separate from synthetics — different processing streams entirely

Damage-specific steps — pulled from ConditionScore tags

- **Stained items:** note stain type if known — oil-based stains can contaminate a whole batch
- **Mold/odor:** flagged as a hard reject before the user travels to drop it off

4.4 Guided Walkthrough

1

GLIE returns a RECYCLE verdict

GLIE score falls below 0.565. No extra user action needed — this is the existing assessment flow.

2

Checklist auto-builds from existing data

Universal steps load by default; fiber- and damage-specific steps pull from T2 and ConditionScore automatically.

3

User confirms each item

Checkbox-style, actively ticked — a behavioural nudge rather than a disclaimer nobody reads.

4

Hard-reject items caught early

Mold/odor flags surface before travel, not on arrival.

5

Proceeds to RVRT facility match

Flow continues exactly as already specified — RVRT selects a verified partner.

4.5 Screen Reference

PREP THIS ITEM FOR RECYCLING

[x] Wash & dry completely

[x] Remove zippers / buttons

! Mixed fiber (cotton-lycra) — keep separate from pure cotton

I'VE PREPPED IT → FIND DROP-OFF

4.6 Technical Implementation

Extends the existing **POST /api/glue/assess** RECYCLE response — same endpoint RVRT already extends. No new AI call required: rules-based lookup against data already produced.

RECYCLE response additions:

```
{
  ...existing GLIE/RVRT fields,
  "prep_checklist": {
    "universal": [...],
    "fiber_specific": [...], // derived from T2 lookup
    "damage_specific": [...] // derived from ConditionScore tags
  },
  "hard_reject_flag": boolean // e.g. mold/odor detected
}
```

Tie-in to RVRT: the `hard_reject_flag` and fiber-mix data can also feed RVRT's facility matching — a mixed-fiber garment shouldn't route to a single-fiber-only facility. Natural Phase 2 extension.

PART 5

Full Feature Roadmap

Every feature idea discussed for the recycling flow, grouped by build priority

5.1 Already Designed — Just Needs Building

- **Verified Facility Tier (RVRT)** — route RECYCLE-branch garments only to partners with documented PPE/ventilation/no-open-drain criteria, starting with the 5 verified-address partners in Part 3
- **RouteScore-based facility selection** — weight facility trust higher for garments needing harsher processing, so the riskiest items go to the safest facilities
- **Labour-condition badge** at the disposal screen, next to the existing carbon badge
- **180-day verification decay** — auto-downgrade a facility's trust tier if not re-checked

5.2 New — Grounded in the Research

- **Contamination & sorting guidance** — full spec in Part 4
- **Drop-off locator** — map view of verified partners near the user, with distance, accepted materials, and verification tier shown before commitment
- **Source-aware routing toggle** — let users opt into “domestic-only” partners, avoiding import-heavy hubs like Panipat by default
- **Pre-consumer vs. post-consumer framing** in eco-impact copy — position KaPhor as targeting the specific post-consumer gap rather than implying all textile waste is equally hard to recycle
- **“Where it actually goes” transparency card** — a short explainer per material type (e.g. mechanical recycling shortens cotton fibers ~30–40%, hence RESELL/UPCYCLE tried first) so “recycled” claims aren't made without context
- **Self-onboarding form for new partners** — lets facilities with unverified addresses (Bombay Recycling Concern etc.) submit details to populate T6 at SELF_REPORTED tier
- **Waste-picker/informal-sector acknowledgment** — a simple “this process supports X livelihoods” note, tying into the 4–4.5 million livelihoods stat and the “job quality, not just job creation” framing

5.3 Lower Priority / Later Phase

- CV-based photo verification of facility safety conditions — deferred per RVRT's own next-steps; no labelled data yet
- Automated contamination detection directly from the garment photo already captured at GLIE assessment time

5.4 Why Contamination Guidance Ships Before More Facility Partners

Reason	Why it matters
Cheapest to ship	Rules-based, reuses existing GLIE and T2 outputs, adds one new screen — no new AI call, no new data pipeline
Targets the actual bottleneck	The Ministry of Textiles report names contamination and segregation — not facility count — as the binding constraint

Reduces wasted trips	Hard-reject flags (mold, odor) are caught before the user travels anywhere
Strengthens RVRT downstream	Better-prepped, correctly fiber-tagged garments make RVRT's facility matching more accurate without changing RVRT itself

PART 6

Caveats & Open Data Gaps

What to flag before citing any of this formally

- No source provides a verified, government-published count of textile recycling centers in India. Facility-level figures (e.g. Panipat's unit counts) come from journalism and industry blogs, not regulatory filings, and disagree by up to two orders of magnitude depending on source definitions
- Total annual waste-generation estimates range from 1.2 to 8 million tonnes depending on source; the 7.7–7.8 million tonnes figure is used because three independent sources corroborate it, but this is a judgement call, not a settled number
- Textile-specific EPR registration data (parallel to the plastic/e-waste/battery EPR counts that do exist) was not located in this research
- Consumer-facing drop-off networks (Goonj, SADS, etc.) publish city-level coverage but not exact addresses for most locations — only a handful of physical addresses were confirmed, listed in Part 3
- The worker-health findings are well-corroborated across five independent outlets but are specific to Panipat; whether the same severity applies to other, less-documented recycling clusters in India is not established by any source reviewed
- All web sources for Part 1 and Part 3 were accessed June–September 2026 and should be cross-checked against primary sources before formal academic citation

KaPhor — AI-Driven Circular Fashion Platform | Recycling Feature Master Reference | Consolidates the Textile Recycling Briefing, RVRT spec, verified partner research, and Contamination & Sorting Guidance